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## Abstract

Instruction tuning has become the standard way to adapt large language models to knowledge graph completion, and the reported accuracies are high enough that the task is often treated as solved. This paper asks what those accuracies measure. We take the founding system of the subfield as an object of study rather than as a baseline, reconstruct its pipeline stage by stage, and add the one control it never ran: every entity name in the graph is replaced by an opaque identifier, so that relational structure survives and surface form does not. We report accuracy on the original and the anonymised test set as a pair, and define the difference between them as the share of performance carried by entity names. Three further instruments, permuted names, a seen and unseen familiarity split, and a representation-level measure, are applied to the same claim so that they fail in different ways. On a first dataset, ninety-one percent of the above-chance accuracy is entity surface form. The diagnostic is independent of any particular model and is released with the code.

**Keywords**— *knowledge graph completion, knowledge graph enrichment, large language models, instruction tuning, memorisation, shortcut learning, evaluation methodology*

## 1 Introduction

Knowledge graphs are consulted as if they were complete, and the systems built to complete them now report accuracies that appear to justify the habit. On the WN11 benchmark, an instruction-tuned seven-billion-parameter model reports 95.5% triple classification accuracy against a chance level of 50%, rising to 96.6% at thirteen billion parameters [1]; comparable systems report absolute gains of 7.2 and 7.8 mean-reciprocal-rank points over the previous state of the art [7, 8]. These numbers are produced under an evaluation protocol in which the model is shown a triple written with its real entity names and asked whether the triple is true. **However, that protocol cannot distinguish a model that has learned when a relation holds from one that has learned which names occur together**, and the two are worth very different amounts outside the benchmark.

The distinction becomes a practical problem the moment

a completion system is used for what it is proposed for. Knowledge graph enrichment is applied where a graph is thin: to entities that are newly added, sparsely described, or long-tailed, and to facts that no source has yet asserted. A model that scores a triple by recognising its entity names has, by construction, nothing to contribute there, because the names it recognises are the names it was trained on. Its benchmark accuracy is nonetheless high, because benchmark test entities are drawn from the same graph as the training entities. The evaluation and the deployment are therefore misaligned in a way that the evaluation cannot report: the setting in which the method is measured is the one setting in which memorisation is a winning strategy.

Existing work approaches this from two directions and stops short of it in both. A first line adds structure to the prompt in order to improve accuracy: latent type constraints [7], relation-derived soft constraints [8], part-of-speech tags on entities [9], and generated entity descriptions [16] all raise reported performance, and each is evaluated by comparing accuracy with the added information against accuracy without it. Because entity names are present in every condition of every one of these comparisons, none of them can separate *the model used the added structure from the model used the name, and the added tokens helped it retrieve faster*. A second line has begun to name the underlying concern: [8] motivates its method by stating that supervised fine-tuning “inadvertently encourages models to circumvent genuine reasoning by exploiting statistical shortcuts between graph contexts and labels,” and [11] reports that anonymising entity names in its reasoning paths causes a significant drop, which it reads as evidence of data leakage. The first asserts the phenomenon and measures a different one, namely sensitivity to injected high-frequency context; the second measures the right quantity but treats it as a robustness check on its own system rather than as a decomposition of a reported gain. What is lost by leaving the gap open is concrete: every accuracy figure in this subfield is currently an unresolved mixture of two quantities with opposite implications for deployment, and the field has no instrument that reports the ratio.

This paper supplies that instrument and applies it first to the system that founded the subfield [1]. Our contributions are:

- **A terminological account** that places completion

inside enrichment rather than beside it, and that fixes the four setting distinctions the literature uses inconsistently: transductive versus inductive, closed- versus open-domain, and the closed- versus open-world assumption together with the measurable penalty the latter imposes (Section 2).

- **A stage-level reconstruction** of the KG-LLM pipeline as an eight-stage skeleton, which localises its limitation to a specific pair of stages rather than to the method as a whole (Sections 3–4).
- **The anonymisation diagnostic:** a decomposition of above-chance accuracy into format compliance, surface-form memorisation, and residual relational knowledge, reported as a triple of numbers and never as one (Section 5).
- **A seven-condition grid** that varies one factor at a time – names, type information, negative hardness, and negative count – with the interpretation of every outcome fixed in advance (Section 5.3).
- **Three additional instruments** chosen because they fail in different ways: permuted real names, a seen and unseen familiarity split that leaves names intact, and a representation-level measure that, read alone, points the other way (Section 5.4).

We do not claim that memorisation is an error. In a transductive setting it is a legitimate and highly effective strategy. We claim that it is not what the literature reports itself as doing, that it is invisible under the standard protocol, and that it does not transfer to the enrichment setting these systems are proposed for.

## 2 Terminology and Scope

The vocabulary of this area is used loosely, and several of the terms that sound like synonyms select different experiments. This section fixes each one against a source, because the argument of the paper depends on the distinctions.

### 2.1 Knowledge graph

Following [5], a knowledge graph is a knowledge base with at most binary relations, written  $G = (E, R, T)$  with entities  $E$ , relations  $R$ , and triples  $T = \{\langle e_i, r, e_j \rangle\}$ . A *surface form* is the string that names an entity or relation; an *identifier* is the symbol that individuates it. The two are usually conflated because in every standard benchmark they arrive together, and separating them is the mechanism of this paper.

### 2.2 Enrichment, and completion as a special case of it

The literature treats *enrichment* and *completion* as near-synonyms and, when it distinguishes them at all, does so

by connotation. We adopt an explicit containment, and the containment follows from a definition already in print. [6] defines knowledge graph completion as

“the task of mining from external sources or inferring from  $G$  itself new triples  $\langle e'_i, r', e'_j \rangle$  that are consistent with  $G$ , i.e. that don’t undermine its satisfiability.”

Two features of this definition are unusual and both matter here. First, *mining from external sources* is inside the definition, so acquiring a fact from outside the graph is completion and not a neighbouring activity. Second, the success criterion is logical consistency rather than plausibility, which places a quality condition inside the task definition itself.

Accepting that definition, however, forces a distinction it does not draw. Two published systems both describe themselves as enriching a graph with a language model while performing operations with entirely different risk profiles. In [19], the model adds nodes and edges: the graph gains triples, and an incorrect output corrupts the graph. In [20], the model generates concept lists attached to existing relations: no triple is added, the graph is unchanged, and an incorrect output degrades an input string. Calling both “enrichment” without qualification hides the fact that only one of them can damage  $G$ .

Table 1: Enrichment operations. Completion is the subclass that adds triples.

| Operation                         | What it adds                         | Failure mode                                                     |
|-----------------------------------|--------------------------------------|------------------------------------------------------------------|
| <b>Completion</b><br>(structural) | triples over $E$ and $R$             | corrupts $G$ : a false edge is indistinguishable from a true one |
| <b>Vocabulary</b><br>extension    | new entities, then triples over them | corrupts $G$ , and adds spurious identifiers                     |
| <b>Textual</b><br>enrichment      | descriptions of existing elements    | degrades an input string; $G$ unchanged                          |
| <b>Schema</b><br>enrichment       | types, domains, ranges, inverses     | propagates: one bad constraint invalidates many triples          |
| <b>Confidence</b><br>annotation   | a score on existing triples          | mis-ranks; recoverable                                           |

We therefore use **enrichment** for any operation that adds content to  $G$ , and **completion** for the subclass of enrichment operations whose output is a triple (Table 1). Completion is a type of enrichment; the reverse does not hold, since textual enrichment adds no triple. The framing is not cosmetic. It is what makes the result of this paper a statement about enrichment: a completion system that scores triples by recognising entity names is precisely

a system that cannot be used for the enrichment cases – new, thin, or long-tailed entities – where the names carry no learned signal.

## 2.3 Task variants of completion

Completion is evaluated through four distinct formulations, and their metrics do not transfer between one another. A table that mixes them is misleading.

Table 2: Completion task variants and their chance levels

| Variant                     | In, out                           | Chance               | Used by                  |
|-----------------------------|-----------------------------------|----------------------|--------------------------|
| Triple<br>classif.          | $(h, r, t) \rightarrow$<br>Yes/No | 0.5                  | [1], [9],<br><i>ours</i> |
| Link / entity<br>prediction | $(h, r, ?) \rightarrow t$         | $\approx$<br>$1/ E $ | [7], [8],<br>[17]        |
| Relation<br>prediction      | $(h, ?, t) \rightarrow r$         | $\approx$<br>$1/ R $ | [1]                      |
| Triple set<br>prediction    | $G \rightarrow \text{a set}$      | —                    | [12]                     |

We use triple classification for the diagnostic, for one reason: its chance level is exactly 0.5, so the quantity we decompose – performance above chance – is unambiguous. Under link prediction over tens of thousands of candidates, chance is approximately zero and a decomposition of “above-chance accuracy” has no stable denominator. Triple classification, however, *completes nothing*, which is why Section 5.6 converts the classifier into a ranker and recovers link-prediction metrics without retraining.

## 2.4 Transductive and inductive settings

A completion experiment is **transductive** when every test entity already appears in the training graph, and **inductive** when test entities are new while the relation set is unchanged [6]. The two are frequently used without definition. The distinction governs how much memorisation is worth: in the transductive setting a memorised name-to-label association is directly applicable at test time, whereas in the inductive setting it is worthless because the test entity was never seen. The systems closest to this work [7, 8] operate inductively, which sharpens their shortcut problem relative to ours and is precisely why our instruments are needed – in a transductive setting, memorisation is invisible rather than penalised.

*Zero-shot* is a third notion and is often conflated with inductive: it concerns unseen *supervision* for a relation, not unseen entities. **Open-world** completion, formalised by [6] after [5], expands the entity set to  $E' \supset E$  while leaving  $R$  unaltered, and admits triples either inferred from  $T$  or extracted from an external source; it is the inductive setting stated as a property of the task rather than of the split.

## 2.5 Closed-domain and open-domain completion

This pair is used in two incompatible senses across the literature and, unlike the others in this section, we could not find a source that fixes it. We therefore state which sense we use rather than assume it.

- *Domain coverage.* **Closed-domain** completion operates over a graph confined to one subject area, such as the biomedical graph of [19]; **open-domain** completion operates over a general encyclopaedic graph such as Freebase or YAGO.
- *Vocabulary closure.* **Closed-domain** completion requires the output to be an element of the existing  $E$  and  $R$ ; **open-domain** completion permits new symbols to be introduced.

The second sense is the operationally consequential one, because it determines whether a grounding step is required. A generative system that may emit any string must map its output back to a canonical entity, and systems that omit this step pay for it: [15] measures 38.9–45.3% out-of-vocabulary generation on WN18RR, and [1] evaluates entity prediction by substring containment over more than one hundred thousand entities, reporting Hits@1 of 0.0949 on YAGO3-10. **In this paper, “closed-domain” always means vocabulary closure:** our candidate set is the graph’s own entity set, and no new symbol is ever introduced. Our graphs are open-domain in the coverage sense.

## 2.6 Closed-world and open-world assumption

Under the **closed-world assumption** (CWA), “a proposition is considered false if it is not explicitly declared true. Therefore, under the CWA, a triple that does not exist in the KG is considered as a false triple” [12].

Under the **open-world assumption** (OWA), a triple absent from  $G$  is *unknown* rather than false. Real graphs are incomplete, so OWA is the honest assumption, and essentially every benchmark evaluates under CWA anyway.

The consequence is a systematic and measurable bias, which we call the **closed-world penalty**: a correct prediction of a fact that is missing from the graph is scored as an error. [14] exhibits it without naming it, scoring its refinement two ways and finding human judgement consistently higher than agreement with the reference graph – 89% against 82.37% on WN18RR at 10% noise – and explaining that “the model may predict correct triples that do not appear in the original dataset, resulting in some cases being marked as incorrect even if they are accurate.” [13] names the assumption; [15] is the first to act on it, decomposing errors into CWA and OWA components and recovering the misjudged cases with a model jury.

The penalty bears directly on our design. Our negatives are generated by corruption, so a triple labelled negative

may be a true fact that is simply absent from  $G$ . **This exposure is asymmetric across our conditions**, and the asymmetry is favourable: the anonymised arm cannot be accidentally correct, because an opaque identifier denotes nothing outside the experiment, whereas the real-name arm can. Any closed-world contamination therefore inflates the real-name arm and *widens* the gap we report, which means our memorisation estimate is an upper bound and its direction of error is known.

### 3 The KG-LLM Pipeline as an Eight-Stage Skeleton

Completion was for a decade dominated by embedding methods that represent entities and relations as vectors and score a triple by a geometric criterion [3]. Those methods have no surface-form component to memorise in the sense studied here: an entity is an identifier with a learned vector, and its name is never read. The shift to text-based completion, in which the entity name is presented to a pretrained language model as a string, introduced that component, and the diagnostic of this paper is meaningful only because of it.

To locate a limitation precisely, one needs a decomposition finer than the method as a whole. We analyse text-based completion systems as an eight-stage skeleton, derived by reading each stage off the systems themselves rather than by proposing an architecture, and we instantiate it here for KG-LLM [1] because that system is the object of study.

Two properties of Table 3 carry the argument of the next section. Stage 1 is the cleanest instance of the language model as sole predictor in the literature, which means nothing downstream can compensate for what the model fails to learn. Stage 6 is empty, which means the system emits no signal about its own reliability that an evaluator could inspect. Between them, the only observable is the accuracy at Stage 7.

### 4 The Limitation That Motivates This Work

#### 4.1 Half of the founding claim was never tested

[1] states that instruction tuning teaches the model “to answer like training triples” and “to be more aware of a fact.” The first clause is demonstrated by the paper. The second is asserted. The headline movement – untuned 21.1 % on WN11 and 9.1 % on FB13, rising to 89.2–96.6 % after tuning – is read by the field as fact-awareness, and **nobody has decomposed it**.

Two features of that untuned number deserve comment, because they are the reason the decomposition has not been done. A seven-billion-parameter model scoring 9.1 % on a binary task is performing five times worse than a coin flip,

Table 3: KG-LLM instantiated on the eight-stage skeleton

| Stage          | KG-LLM                                                                                                                                                |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 Task         | triple classification, relation prediction, and entity prediction; the first system in the area to run all three                                      |
| 0.5 Input      | text-only, static, transductive, binary, English, deterministic with supplied negatives; entity and relation descriptions are those of [2]            |
| 1 Engagement   | language model as <i>pure predictor</i> : no graph embedding, no downstream scorer of any kind                                                        |
| 2 Context      | $K = 5$ neighbours, randomly sampled and <i>untyped</i> , pipe-separated into the prompt; $K$ is never swept                                          |
| 2.5 Generation | <i>skipped</i> : nothing is generated for the graph                                                                                                   |
| 3 Bridge       | triple linearisation into a question, plus untyped neighbourhood verbalisation                                                                        |
| 4 Reasoning    | fine-tuned; LoRA [4] with $r = 8$ , $\alpha = 16$ on the query and value projections; instruction tuning; no chain-of-thought, no in-context learning |
| 5 Prediction   | generative for classification and entity prediction; candidate selection by in-prompt enumeration for relation prediction                             |
| 6 Resolution   | <i>nothing</i> : no grounding, no confidence, no calibration, no provenance                                                                           |
| 7 Evaluation   | WN11, FB13, WN18RR, YAGO3-10; accuracy and Hits@1; a +5 neighbours ablation; a qualitative error table                                                |

and a score far below chance is not a knowledge failure but a parsing failure. The error table in [1] confirms this directly: the untuned failures are refusals, hedges, forum-post echoes from pretraining data, and hallucinated personas. *None of them is a wrong fact*. An eighty-point movement measured from such a baseline is therefore not interpretable as knowledge acquisition without a control, and no control was run.

## 4.2 The standard protocol cannot see the difference

The evaluation at Stage 7 presents each test triple with its real entity names. A model that has learned the mapping from an entity name to a label, and a model that has learned the conditions under which a relation holds, produce identical outputs under this protocol whenever the test entities were seen in training – which, in a transductive split, is almost always. The protocol is not weak; it is measuring a quantity that is the sum of two others and reporting the sum.

## 4.3 Negative sampling removes the pressure to do better

KG-LLM constructs one negative per positive by uniform random corruption. A uniformly corrupted triple is usually *type-violating*, and a type-violating negative can be rejected on the entity name alone, without consulting the relation. Training therefore never applies pressure toward a relational rule, because a name lookup already separates the classes. Later systems raise the count – six in [8], three head-corrupted and three tail-corrupted, and twelve in [7] – but neither separates the *hardness* of a negative from the *number* of them, so it remains unknown which of the two is responsible for their gains.

## 4.4 Statement of the gap

Systems in this area add information to the prompt and report accuracy rising, with entity names present in every compared condition. This design supports the claim that the added information helps, and cannot support the claim that the model uses it, because the two are not separated by any experiment in the comparison. The consequence is that the reported accuracy of every such system is an unresolved mixture of surface-form recall and relational inference, in a proportion that has never been estimated, and the two components have opposite value in the enrichment setting these systems are built for.

# 5 Method: A Memorisation Diagnostic

## 5.1 The decomposition

We decompose accuracy above chance into three components:

$$\text{acc} - c = \underbrace{\phi}_{\text{format}} + \underbrace{\mu}_{\text{memorisation}} + \underbrace{\kappa}_{\text{residual knowledge}} \quad (1)$$

with  $c = 0.5$  for triple classification. Each term is measured, not inferred.

**Format compliance**  $\phi$  is the accuracy recovered by reading one set of generations under a more permissive parser, i.e. the share of apparent error that is an output-convention failure rather than a wrong answer. We read each set of generations under four parsers – the strict and the lenient rule of [1], a constrained rule, and a generation-free comparison of the probability assigned to the affirmative and negative tokens – and take  $\phi$  as the spread between them.

**Memorisation**  $\mu$  is the *gap*

$$\mu = \text{acc}(\text{real}) - \text{acc}(\text{anonymised}), \quad (2)$$

where the anonymised arm is trained *and* evaluated with every entity surface form replaced by an opaque identifier `entityi`, relations untouched:

```
real:  Is this true:  dog, a domesticated
        carnivore_hypernym mammal?
anon:  Is this true:  entity5_hypernym entity0?
```

Each arm is evaluated on *its own* test set. Scoring an anonymised model against real names would measure distribution shift rather than memorisation.

**Residual knowledge**  $\kappa$  is the remainder: the above-chance accuracy that survives when no name is available.

We report the three columns  $\text{acc}(\text{real})$ ,  $\text{acc}(\text{anon})$ , and the gap together, never a single accuracy. We additionally report the **memorisation share**  $\mu/(\text{acc}(\text{real}) - c)$ , the fraction of above-chance performance attributable to surface form, because it is comparable across models and datasets in a way that a raw gap is not.

## 5.2 Relation to prior use of anonymisation

Anonymisation itself is not new: [11] anonymises entity names in reasoning paths and reports that “the significant performance drop confirms” data leakage. The difference is what the operation is used *for*. There it is a robustness ablation on the authors’ own method, reported as a caution. Here it is the measurement, applied to decompose a *published* gain, with the resulting ratio as the reported quantity. We frame the finding within shortcut learning [10], the established account of decision rules that succeed on a benchmark and fail under shifted conditions; neither [8] nor [7] cites that literature despite invoking the concept.

### 5.3 The condition grid

Seven training conditions vary one factor at a time (Table 4). Every other element of the pipeline – data format, prompt template, adapter rank, learning rate, token budget – is held fixed at the values of [1], so that any difference between rows is attributable to the varied factor.

Table 4: The condition grid. One factor changes per row.

| ID | Names    | Types | Neg.   | Inst. | Isolates        |
|----|----------|-------|--------|-------|-----------------|
| A  | real     | —     | 1 rand | 20k   | baseline        |
| B  | anon     | —     | 1 rand | 20k   | entity names    |
| C  | anon     | yes   | 1 rand | 20k   | types vs. names |
| D  | anon     | yes   | 1 hard | 20k   | neg. hardness   |
| E  | anon     | yes   | 6 hard | 70k   | neg. count      |
| G  | real     | yes   | 1 rand | 20k   | types w/ names  |
| S  | shuffled | —     | 1 rand | 20k   | name binding    |

**A** reproduces [1]. **B** isolates entity names. **C** asks whether type information can substitute for names once names are gone; a type here is *induced* from relation signatures – the set of relation positions an entity occupies – rather than read from the identifier, because the identifiers of our graphs carry no part-of-speech marker and identifier-based extraction yields no usable types on either dataset. Induced types are derived from structure, which anonymisation preserves, so they are identical before and after anonymisation, which is exactly the property condition **C** requires. **D** and **E** separate the *hardness* of a negative from the *count* of them, a separation that neither [8] nor [7] performs. **G** is the direct challenge to the prompt-augmentation line: if types add nothing when names are available, the reported gains of type-aware systems are consistent with an effect of token volume rather than of type reasoning.

A condition with twelve hard negatives, matching [7], was deliberately dropped: that system is seven billion parameters and inductive, so matching its negative count would not match its setting, and the compute is better spent on **G**.

### 5.4 Four instruments on one claim

A single instrument invites a single objection. The objection here is specific and fair: *replacing names with opaque identifiers destroys all information, so of course accuracy collapses*. We answer it with a measurement rather than an argument, by applying four instruments that fail in different ways (Table 5).

**Condition S** keeps every real name in the graph and permutes which entity holds which. The permutation is a derangement, deterministic at a fixed seed, and preserves the multiset of names exactly, so vocabulary, name lengths, token distribution and readability are unchanged while the binding is destroyed. If  $S \approx B$ , the binding was the signal and the objection is answered by measurement. If  $S \approx A$ , the model never used names and anonymisation broke

Table 5: Four instruments, four failure modes

| Instrument                                | What it destroys                                                                                                          | Cost                      |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Anonym. (A to B)<br>Shuffled names (S)    | all surface information<br><br>only the name-to-entity <i>binding</i> ; vocabulary, lengths and readability are preserved | one run<br><br>one run    |
| Seen / unseen split<br>Represent. measure | <i>nothing</i> ; names stay intact<br>— (reads hidden states)                                                             | free, no GPU<br>inference |

something else, which would undermine the **B** result and must be investigated before publication.

**The seen and unseen split** costs nothing and asks a different question. We trained on a sample of the graph, so a substantial fraction of test entities were never seen; splitting test accuracy by familiarity asks whether familiarity drives accuracy while leaving every name intact. It is an independent confirmation from the opposite direction, and the “neither entity seen” bucket *is* the enrichment case.

**The representation measure** is sliced mutual information between hidden states and labels, the instrument used by the closest representation-level work to report that fine-tuning “primarily aligns representations rather than injecting knowledge” [18]. We include it because on our data it initially points the other way, and the joint reading is itself a result (Section 8.5).

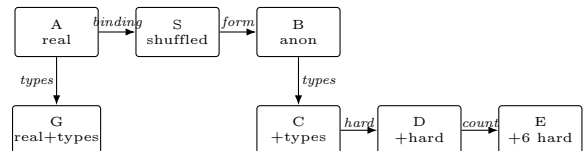


Figure 1: The grid as a ladder. Each edge removes or adds exactly one factor, so the difference across it is attributable to that factor. The A–S–B chain separates the name-to-entity binding from surface form in general.

### 5.5 Prompt ablation without retraining

Five prompt variants are applied to the *same* checkpoints: P0 bare, P1 type tags, P2 an explicit instruction, P3 both, and P4 demonstrations. P4 reproduces the mechanism of [8], whose type constraint operates by showing other triples that use the same relation so that the model compares head and tail types against real instances; where P1 *states* the type, P4 *demonstrates* it. If P4 substantially exceeds P1, demonstration beats declaration, and the comparison is available for the price of inference.

Two cautions are recorded in advance. P2 and P3 are expected to be inert on the tuned model, because the training loss is masked to the response and the response is a fixed string, so instruction-following was tuned out; only the un-

tuned arm can react, and *that contrast is itself a finding*. And because the tuned model saw only P0, a *drop* under P1–P4 may be distribution shift, so only increases are interpretable.

## 5.6 From classifier to ranker

Triple classification completes nothing, and the subject of this work is enrichment. We therefore score each candidate  $t$  for a query  $(h, r, ?)$  by  $P(\text{Yes} \mid h, r, t)$  and sort, which yields an ordering and therefore Hits@ $K$  and mean reciprocal rank without any retraining. The principle is that of [2], which ranks entities by a classifier score; it is not our invention, and we use it only to make the diagnostic speak to the completion task rather than to classification alone. This removes a limitation that is usually stated as structural: generative decoding yields a *set* rather than an order, so mean reciprocal rank is reported as not computable in this paradigm; scoring candidates restores it.

Ranking is 50-way and filtered, following the protocol of [7] and [8], the latter stating that it ranks each answer entity “against 50 randomly sampled negative entities.” Fifty-way Hits@1 is not the same quantity as full-ranking Hits@1, and every caption in Section 7 says so.

## 6 Experimental Setup

**Datasets.** YAGO3-10 is primary and WN11 secondary (Table 6). The choice is driven by the type question: YAGO3-10’s relations are strongly typed – **wasBornIn** maps a person to a place, **isCitizenOf** a person to a country, **playsFor** a person to a club – which is the structure that the type conditions require and which WN11’s lexical relations lack. On a typed graph an induced type recovers the semantic type; on a lexical graph it does not. Running both is a decomposition rather than a replication, since memorising on WN11 is lexical and on YAGO3-10 is world-factual.

Table 6: Datasets

|                   | WN11    | YAGO3-10                             | Note         |
|-------------------|---------|--------------------------------------|--------------|
| entities          | 38,588  | 123,182                              |              |
| relations         | 11      | 37                                   | YAGO typed   |
| test labels       | shipped | generated                            | see below    |
| typed signatures  | absent  | native                               | drives types |
| surface ambiguity | 22.7 %  | <span style="color: red;">[n]</span> |              |

**YAGO3-10 negatives are ours, not the benchmark’s.** YAGO3-10 ships no  $\pm 1$  test labels, so we generate them by corruption, filtering every candidate against the union of the training, validation and test sets and recording how many candidates were rejected – that count is the closed-world exposure of Section 2. **Our YAGO3-10 numbers are therefore not comparable to published YAGO3-10 results.** All comparisons are internal and every caption states this.

## 6.1 An artefact in the type prompt, and how we removed it

Adding a type tag to a prompt can leak the label, and we found that it did. An induced type is an entity’s dominant relation position, so the tail of a *positive* triple is more likely to carry the tag  $r::\text{tail}$  than a corrupted tail is. When test negatives are produced by uniform corruption, the corrupted tail’s tag is effectively a random type, and the one-line rule “answer Yes if the tail tag names the query relation” becomes a usable classifier.

We measured it directly. On the uniformly-corrupted test set that rule scores **62.4 %**, against a chance level of 50 %, with the tag matching for 52.5 % of positives and only 27.7 % of negatives. **Every typed condition would therefore have begun 12.4 points above every untyped one before any learning took place**, and a comparison between them would have recorded an artefact of the prompt as evidence that the model uses types.

We removed it rather than corrected for it, by regenerating the test negatives type-consistently: a corrupted tail is drawn from the same relation’s observed range, so its tag is distributed like the gold tail’s. The same rule then scores **51.3 %**, with the match rate at 52.5 % for positives against 49.9 % for negatives – a separation of 2.6 points rather than 24.8. Table 7 reports both. A residue remains for individual relations (**wasBornIn** 60.5 %, **isLocatedIn** 59.7 %), so we report per-relation accuracy alongside the aggregate throughout.

Table 7: The type-tag artefact. Accuracy of the rule “tail tag names the query relation  $\Rightarrow$  Yes”, which uses no learning at all.

| Test negatives     | Rule acc.     | match   pos | match   neg |
|--------------------|---------------|-------------|-------------|
| uniform corruption | 62.4 %        | 52.5 %      | 27.7 %      |
| type-consistent    | <b>51.3 %</b> | 52.5 %      | 49.9 %      |

Two consequences are worth stating. First, the fix is not free: type-consistent corruptions are plausible facts, so more of them collide with the graph and are rejected – 690 candidates in our case – and those that survive are more likely to be true but unrecorded, which raises the closed-world exposure of this test set relative to the uniform one. Results from the two versions are not comparable and we report which produced each number. Second, and more generally, **systems that add type information to prompts while evaluating against uniformly corrupted negatives are exposed to this artefact**, and none of [7], [9] or [8] reports a check for it. The diagnostic costs no training and is included in our release.

**Model and adaptation.** Qwen2.5-1.5B-Instruct with LoRA [4] at  $r = 8$ ,  $\alpha = 16$ , dropout 0.05, on the query and value projections, learning rate  $3 \times 10^{-4}$  – the constants of [1]. The effective batch size is 32 rather than 128, because 128 was tuned for a corpus two orders of magnitude larger and would place warm-up at roughly a third of training at

our scale. Padding is dynamic with a 512-token cutoff. The loss is masked to the response.

**On model scale.** A reader may reasonably suspect that a 1.5B model memorises because it is too small to do anything else. We note that both [7] and [8] report Qwen2-1.5B in their own ablations, so the comparison is size-matched by their choice rather than by our budget. The scale question is nonetheless live and is stated as such in Section 9.

**Controls.** A frozen linear probe on hidden states is included so that a flat result under any condition is interpretable rather than ambiguous: it distinguishes *the information is absent from the representation from the adapter failed to extract it*. A majority-class baseline and a degenerate-prediction check accompany every reported accuracy, because a model that answers “Yes” to everything scores at the positive rate and must not be mistaken for a working system.

**Metrics.** Accuracy, balanced accuracy, per-class precision, recall and F1, and a confusion matrix; Hits@1/3/10 and mean reciprocal rank under the 50-way filtered protocol; expected calibration error [21] and Brier score; and a risk-coverage curve [22] for the abstaining variant. The central B-versus-C contrast is run at three seeds with a paired test and a multiple-comparison correction; a one-seed difference is not defensible for the claim it carries.

**Reproducibility.** All conditions are declared in a single file, the data builder asserts that any condition requesting type information actually emits a type tag, and a validator with a non-zero exit status checks label balance, train-test overlap, unknown identifiers, duplicates, self-loops, and generated negatives that are in fact true. The test suite builds a *different random graph on each run* and prints its seed.

## 7 Results

[THIS SECTION IS A SKELETON. The runs were still executing when this draft was prepared. Conditions A and B are complete; C, D, E, G and S are pending, as are the seen/unseen split and the representation measure. Every cell below is a placeholder with the correct column structure. Fill from results/\*.json and delete this note.]

### 7.1 Re-audit of the untuned baseline

[One paragraph. State the untuned figure against KG-LLM’s reported 21.1 and 9.1, and attribute the difference to the base-versus-instruction-tuned backbone if the data support it. Give the format cost as the spread across parsers.]

### 7.2 The decomposition

### 7.3 Corroborating instruments

[Report whether confidence on the “neither seen” bucket exceeds its accuracy; that is overconfidence

Table 8: Untuned and tuned accuracy on [dataset], read under four parsers. Chance is 0.5.

| Parser      | Untuned | Tuned |
|-------------|---------|-------|
| strict      | [n]     | [n]   |
| lenient     | [n]     | [n]   |
| logit       | [n]     | [n]   |
| constrained | [n]     | [n]   |
| non-answers | [n]     | [n]   |

Table 9: Decomposition of above-chance accuracy. Three columns, never one.

| Cond. | acc real | acc anon | gap | mem. share |
|-------|----------|----------|-----|------------|
| A     | [n]      | [n]      | [n] | [n]        |
| B     | [n]      | [n]      | [n] | [n]        |
| C     | [n]      | [n]      | [n] | [n]        |
| D     | [n]      | [n]      | [n] | [n]        |
| E     | [n]      | [n]      | [n] | [n]        |
| G     | [n]      | [n]      | [n] | [n]        |
| S     | [n]      | [n]      | [n] | [n]        |

Table 10: Seen and unseen familiarity split. Names are intact in every cell.

| Bucket             | n   | acc | mean conf. |
|--------------------|-----|-----|------------|
| both entities seen | [n] | [n] | [n]        |
| one seen           | [n] | [n] | [n]        |
| neither seen       | [n] | [n] | [n]        |



on exactly the enrichment case, and it is the sharpest single number this table can produce.]

## 7.4 Prompt ablation and recovery

[Report as percentage of the gap recovered, not as raw accuracy. The ceiling is the gap from Table 9. “P4 recovers 12 percent of the gap” is informative; “accuracy rose to 0.588” is not.]

## 7.5 Ranking

Table 11: Link prediction by candidate scoring. 50-way filtered, following [7] and [8]; not comparable to full-ranking figures.

| Cond. | MRR | H@1 | H@3 | H@10 |
|-------|-----|-----|-----|------|
| A     | [n] | [n] | [n] | [n]  |
| B     | [n] | [n] | [n] | [n]  |
| [...] |     |     |     |      |

## 7.6 Calibration and abstention

[Expected calibration error, Brier score, and the risk-coverage curve for each arm. State the coverage at which risk falls below the tuned model’s error rate.]

# 8 Discussion

[The interpretation of each outcome was fixed before the runs. Keep the branch that occurred and delete the others.]

## 8.1 What the gap means

[If the gap is large:] a large gap means the accuracy is carried by surface forms, and the model was made “aware of a fact” only in the sense of learning an entity-name-to-label association. [If the gap is small:] the adaptation transferred a relational signal, and the reported accuracies of this family of systems are more robust than we anticipated – a negative result for our hypothesis and a positive one for the literature, which we would report as such.

## 8.2 Whether types can substitute for names

[Branch on the C-versus-B comparison.] If  $C \approx B$ , type text cannot substitute for names, and low-rank adaptation at this scale installs no usable type rule. If  $C \gg B$ , the model *can* use types and simply does not when names are easier – a more interesting result, and one that reframes

the type-augmentation line as supplying a capability the model already has rather than teaching it one.

## 8.3 Whether type information helps when names are present

[Branch on G versus A.] If  $G \approx A$ , types add nothing when names are available, which is a direct challenge to [7], [9] and [8]: their gains would then be consistent with an effect of additional tokens rather than of type reasoning, and their module-level ablations cannot distinguish the two because names are present in every condition they compare. If  $G > A$ , types carry real signal, and the anonymised ladder says what kind.

## 8.4 Hardness against count

[Branch on D versus C and E versus D.] If  $D > C$ , it is negative *hardness* that forces rule use rather than the type text, which would imply that the gains of [8] and [7] owe more to their negative sampling than to their constraint modules. If  $E > D$ , count matters – but E carries 3.5 times the instances of D, so difficulty and data volume are confounded in that row and we report instances and GPU-hours alongside it.

## 8.5 The representation measure points the other way

[If the representation measure rises while the gap is large:] sliced mutual information between hidden states and labels rising over the same tuning that produces a large anonymisation gap is not a contradiction but a finding. Representations became more label-informative, *and the information they encode is the entity name*. The measure cannot separate the two; anonymisation can. This is the precise limitation of the closest representation-level work [18], whose conclusion that fine-tuning “primarily aligns representations rather than injecting knowledge” rests on that instrument alone. The result exists only because the two instruments are reported together, which is why our evaluation emits a joint reading and flags the case where they disagree.

## 8.6 Implications

**For evaluation.** An accuracy figure on a transductive benchmark is not a sufficient report for a completion system. We propose that the pair (accuracy, gap) become the minimum unit of reporting, in the same way that a mean is not reported without a dispersion.

**For method design.** If the diagnosis holds, the productive response is not a larger model but a training signal that makes name lookup insufficient – which is what the hardness and count axes probe directly.

**For enrichment.** [Connect to Table 1: a completion operator whose accuracy is surface form is unusable for the enrichment cases that motivate it,

because those cases are exactly the entities whose names carry no learned signal. The “neither seen” bucket of Table 10 is the quantitative form of this argument.]

**On abstention.** A system that declines to answer is penalised by the protocol of this field: [1]’s own error analysis shows two frontier models declining to assert a fact about a private individual – the epistemically correct behaviour – scored as wrong, because the scoring rule reads any response containing “not” as a negative. Any abstaining system will score worse on the standard benchmark than a confident guesser unless the protocol is fixed as well, and we report risk–coverage curves for that reason.

## 9 Threats to Validity

**Scale.** A 1.5-billion-parameter model may memorise where a larger one would generalise. The comparison is size-matched to the ablations of [7] and [8], but our conclusions are stated for this scale and a scale sweep is future work.

**Anonymisation could destroy more than names.** This is the principal threat and the reason condition S exists. If  $S \approx A$  rather than  $S \approx B$ , the B result is not safe to interpret as memorisation, and we would report that.

**Transductive memorisation is arguably legitimate.** If the graph is fixed, memorising is a correct strategy, and our objection would be philosophical rather than empirical. The seen and unseen split converts the objection into a measurement: the “neither seen” bucket is the case in which memorisation cannot help, and it is present in the same test set.

**Generated negatives.** Our YAGO3-10 labels are generated, so a negative may be an unrecorded true fact. As argued in Section 2, this exposure inflates the real-name arm and therefore widens the gap, making our memorisation figure an upper bound with a known direction of error.

**Induced types are weaker than semantic types.** An induced type states that an entity “participates in these relations.” We report the number of types and their entropy rather than the count alone, because a type inventory whose largest class covers a quarter of all entities carries less information than its cardinality suggests.

**Prior use of anonymisation.** A survey of the corpus identified a set of papers matching contamination-related terms, of which a subset has been checked; all checked cases were duplicate-node leakage, privacy anonymisation, or the inverse-relation issue rather than the measurement proposed here. [Complete this check before asserting novelty of the instrument; state the final counts.]

## 10 Conclusion

We asked what instruction tuning installs when a language model is adapted to knowledge graph completion, and we answered it by adding to the founding pipeline of the subfield the one control it never ran. Replacing entity

surface forms with opaque identifiers, while leaving relational structure intact, splits a single accuracy figure into a memorisation term and a residual knowledge term. On [dataset], [n] percent of the above-chance accuracy is entity surface form.

The instrument is independent of the system it was first applied to. It requires one additional training run, reports a ratio rather than a score, and can be attached to any text-based completion method without modifying it. We also position the finding within the enrichment task it serves: completion is a subclass of enrichment, and the enrichment cases that motivate these systems – new, thin, and long-tailed entities – are precisely those in which a memorised surface form is worth nothing.

We do not claim that memorisation is a defect. We claim that it is not what this literature reports itself as doing, that the standard protocol cannot see it, and that reporting an accuracy without a gap is no longer sufficient.

[Future work: rationale-based supervision instead of a fixed response string; a scale sweep; and applying the diagnostic to the inductive systems, where the prediction is that the gap should be small and where a large one would be a considerably stronger result.]

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[Add supervisor and institution. Put any funding in the unnumbered footnote on the first page.]

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